

UPI Transaction Analysis — Full Exploratory Data Analysis + Statistical Testing

```
In [1]: # =====  
# CELL 1 – IMPORTS  
# =====  
  
import os  
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import matplotlib.gridspec as gridspec  
import seaborn as sns  
import scipy  
from scipy import stats  
from scipy.stats import (ttest_ind, f_oneway, chi2_contingency,  
                          pearsonr, spearmanr, mannwhitneyu)  
  
import warnings  
warnings.filterwarnings('ignore')  
  
pd.set_option('display.max_columns', None)  
pd.set_option('display.float_format', '{:.4f}'.format)  
sns.set_theme(style='whitegrid', palette='muted')  
  
C = dict(blue='#3B82F6', green='#10B981', red='#EF4444', orange='#F59E0B',  
         purple='#8B5CF6', teal='#14B8A6', pink='#EC4899', dark='#1E293B',  
         bg='#F8FAFC', gray='#6B7280')  
  
# Single output directory used consistently across every cell below  
OUTPUT_DIR = 'outputs'  
os.makedirs(OUTPUT_DIR, exist_ok=True)  
  
print("Imports complete")
```

Imports complete

```
In [2]: # =====  
# CELL 2 – LOAD DATA
```

```
# =====
# NOTE: reads directly from the same folder as this notebook (no 'data/' subfolder)
customer_feedback = pd.read_csv('customer_feedback_surveys.csv')
customer_master = pd.read_csv('customer_master.csv', dtype={'mobile_number': str})
device_info = pd.read_csv('device_info.csv')
fraud_alert = pd.read_csv('fraud_alert_history.csv')
merchant_info = pd.read_csv('merchant_info.csv')
upi_account = pd.read_csv('upi_account_details.csv')
upi_transaction = pd.read_csv('upi_transaction_history.csv')

# — Clean —
upi_transaction['amount'] = pd.to_numeric(upi_transaction['amount'], errors='coerce')
upi_transaction['fraud_flag'] = upi_transaction['fraud_flag'].astype(str).str.upper() == 'TRUE'
upi_transaction['reversal_flag'] = upi_transaction['reversal_flag'].astype(str).str.upper() == 'TRUE'
device_info['is_rooted'] = device_info['is_rooted'].astype(str).str.upper() == 'TRUE'
fraud_alert['resolved'] = fraud_alert['resolved'].astype(str).str.upper() == 'TRUE'
customer_feedback['resolved'] = customer_feedback['resolved'].astype(str).str.upper() == 'TRUE'

print("Tables loaded & cleaned")
for name, df in [('customer_master', customer_master), ('upi_transaction', upi_transaction),
                 ('upi_account', upi_account), ('device_info', device_info),
                 ('merchant_info', merchant_info), ('fraud_alert', fraud_alert),
                 ('customer_feedback', customer_feedback)]:
    print(f" {name:<25}: {df.shape[0]:>7,} rows x {df.shape[1]:>2} cols")
```

Tables loaded & cleaned

customer_master	:	10,000 rows	x	9 cols
upi_transaction	:	100,000 rows	x	15 cols
upi_account	:	12,000 rows	x	6 cols
device_info	:	12,000 rows	x	6 cols
merchant_info	:	500 rows	x	6 cols
fraud_alert	:	2,000 rows	x	7 cols
customer_feedback	:	4,000 rows	x	7 cols

Derived Segmentation Columns: Age_Group & risk_category

```
In [3]: # Age_Group: bucket customer age into brackets
customer_master['Age_Group'] = pd.cut(
    customer_master['age'], bins=[0, 25, 35, 45, 55, 100],
    labels=['18-25', '26-35', '36-45', '46-55', '56+'])
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)

# risk_category: bucket risk_score (0-1) into Low/Medium/High – for customers AND merchants
customer_master['risk_category'] = pd.cut(
    customer_master['risk_score'], bins=[-0.01, 0.33, 0.66, 1.0], labels=['Low', 'Medium', 'High']
)
merchant_info['risk_category'] = pd.cut(
    merchant_info['risk_score'], bins=[-0.01, 0.33, 0.66, 1.0], labels=['Low', 'Medium', 'High']
)

print(customer_master[['customer_id', 'age', 'Age_Group', 'risk_score', 'risk_category']].head())

```

	customer_id	age	Age_Group	risk_score	risk_category
0	cust100000	26	26-35	0.3900	Medium
1	cust100001	49	46-55	0.1300	Low
2	cust100002	68	56+	0.1400	Low
3	cust100003	42	36-45	0.0700	Low
4	cust100004	35	26-35	0.3000	Low

```

In [5]: # =====
# CELL 3 – BUILD RELATIONAL JOINS
# =====

txn_full = (
    upi_transaction
    .merge(customer_master[['customer_id', 'age', 'gender', 'region',
                           'is_business_user', 'risk_score', 'Age_Group', 'risk_category']],
           on='customer_id', how='left')
    .merge(merchant_info[['merchant_id', 'merchant_name', 'merchant_type', 'risk_category']],
           on='merchant_id', how='left', suffixes=('', '_merchant'))
    .merge(device_info[['device_id', 'is_rooted', 'app_version']],
           on='device_id', how='left')
)

# Fraud alert enriched
fraud_enriched = (
    fraud_alert
    .merge(upi_transaction[['transaction_id', 'customer_id', 'amount', 'status',
                           'fraud_flag', 'device_type', 'channel']],
           on='transaction_id', how='left')
    .merge(customer_master[['customer_id', 'region', 'Age_Group', 'risk_score']],

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        on='customer_id', how='left')
    )

    # Feedback enriched
    feedback_full = customer_feedback.merge(
        customer_master[['customer_id', 'age', 'gender', 'region', 'Age_Group', 'risk_score']],
        on='customer_id', how='left'
    )

    print(f"txn_full shape      : {txn_full.shape}")
    print(f"fraud_enriched shape: {fraud_enriched.shape}")
    print(f"feedback_full shape : {feedback_full.shape}")

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txn_full shape      : (100000, 27)
fraud_enriched shape: (2000, 16)
feedback_full shape : (4000, 12)

```

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In [8]: # =====
# CELL 4 – DERIVED COLUMNS (PDF Step 6 requirements)
# =====

# 4A: Transaction Value per Customer = Total Amount / Number of Transactions
customer_txn_stats = (
    txn_full.groupby('customer_id')
    .agg(total_amount=('amount', 'sum'), txn_count=('transaction_id', 'count'))
    .assign(txn_value_per_customer=lambda x: x['total_amount'] / x['txn_count'])
    .reset_index()
)
print("--- Transaction Value per Customer (top 5) ---")
print(customer_txn_stats.sort_values('txn_value_per_customer', ascending=False).head())

# 4B: Merchant Fraud Ratio = Fraud Transactions / Total Transactions by Merchant
merchant_fraud_ratio = (
    txn_full.dropna(subset=['merchant_id'])
    .groupby(['merchant_id', 'merchant_name', 'merchant_type'])
    .agg(total_txns=('transaction_id', 'count'), fraud_txns=('fraud_flag', 'sum'))
    .assign(merchant_fraud_ratio=lambda x: x['fraud_txns'] / x['total_txns'])
    .reset_index().sort_values('merchant_fraud_ratio', ascending=False)
)
print("\n--- Merchant Fraud Ratio (top 5 riskiest, min volume not yet filtered) ---")
print(merchant_fraud_ratio.head())

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# 4C: Device Fraud Ratio = Fraud Transactions / Total Transactions by Device
device_fraud_ratio = (
    txn_full.dropna(subset=['device_id'])
    .groupby(['device_id', 'device_type'])
    .agg(total_txns=('transaction_id', 'count'), fraud_txns=('fraud_flag', 'sum'))
    .assign(device_fraud_ratio=lambda x: x['fraud_txns'] / x['total_txns'])
    .reset_index().sort_values('device_fraud_ratio', ascending=False)
)
print("\n--- Device Fraud Ratio (top 5) ---")
print(device_fraud_ratio.head())

# 4D: Transaction Failure Rate per Customer
customer_failure_rate = (
    txn_full.groupby('customer_id')
    .agg(total=('transaction_id', 'count'), failed=('status', lambda x: (x=='failed').sum()))
    .assign(failure_rate=lambda x: x['failed'] / x['total'])
    .reset_index().sort_values('failure_rate', ascending=False)
)
print("\n--- Transaction Failure Rate per Customer (top 5) ---")
print(customer_failure_rate.head())

# 4E: Global Failure Rate by Channel
channel_failure = (
    txn_full.groupby('channel')
    .agg(total=('transaction_id', 'count'), failed=('status', lambda x: (x=='failed').sum()))
    .assign(failure_rate_pct=lambda x: x['failed']/x['total']*100)
)
print("\n--- Failure Rate by Channel ---")
print(channel_failure)

# 4F: Merchant Type Fraud Ratio (aggregated)
merchant_type_fraud = (
    txn_full.dropna(subset=['merchant_type'])
    .groupby('merchant_type')
    .agg(total_txns=('transaction_id', 'count'), fraud_txns=('fraud_flag', 'sum'))
    .assign(fraud_ratio=lambda x: x['fraud_txns']/x['total_txns'], fraud_pct=lambda x: x['fraud_txns']/x['total_txns']*100)
    .sort_values('fraud_ratio', ascending=False)
)
print("\n--- Merchant Type Fraud Ratio ---")
print(merchant_type_fraud)
```

```
print("\nAll derived columns created")
```

--- Transaction Value per Customer (top 5) ---

	customer_id	total_amount	txn_count	txn_value_per_customer
3933	cust105555	815.0900	4	203.7725
1939	cust102723	550.7500	4	137.6875
6645	cust109461	506.5000	4	126.6250
3983	cust105631	504.4900	4	126.1225
1652	cust102335	468.7000	4	117.1750

--- Merchant Fraud Ratio (top 5 riskiest, min volume not yet filtered) ---

	merchant_id	merchant_name	merchant_type	total_txns	fraud_txns	\
201	merch1201	Garcia, Mann and Sharp	grocery	47	5	
305	merch1305	Ortiz, Gordon and Lutz	electronics	46	4	
358	merch1358	Malone-Willis	electronics	62	5	
80	merch1080	Wilson Inc	food	66	5	
364	merch1364	Taylor LLC	transport	54	4	

	merchant_id	merchant_fraud_ratio
201		0.1064
305		0.0870
358		0.0806
80		0.0758
364		0.0741

--- Device Fraud Ratio (top 5) ---

	device_id	device_type	total_txns	fraud_txns	device_fraud_ratio
7783	dev201118	tablet	1	1	1.0000
7848	dev201223	android	1	1	1.0000
999	dev101401	tablet	1	1	1.0000
394	dev100572	tablet	9	7	0.7778
6104	dev108688	feature_phone	3	2	0.6667

--- Transaction Failure Rate per Customer (top 5) ---

	customer_id	total	failed	failure_rate
3935	cust105559	1	1	1.0000
5071	cust107202	3	2	0.6667
6064	cust108618	5	3	0.6000
6523	cust109286	2	1	0.5000
4226	cust105974	12	6	0.5000

--- Failure Rate by Channel ---

	total	failed	failure_rate_pct
--	-------	--------	------------------

channel			
app	33365	2002	6.0003
intent	33356	1968	5.9000
qr_code	33279	1901	5.7123

--- Merchant Type Fraud Ratio ---

	total_txns	fraud_txns	fraud_ratio	fraud_pct
merchant_type				
electronics	5485	123	0.0224	2.2425
apparel	5986	133	0.0222	2.2219
online	5171	106	0.0205	2.0499
food	4811	95	0.0197	1.9746
transport	4208	83	0.0197	1.9724
grocery	4488	81	0.0180	1.8048

All derived columns created

```
In [6]: # =====
# CELL 5 - EDA FIGURE A: TIME TRENDS
# =====

txn_full['timestamp'] = pd.to_datetime(txn_full['timestamp'])
txn_full['txn_month'] = txn_full['timestamp'].dt.to_period('M').astype(str)

monthly = txn_full.groupby('txn_month').agg(
    txn_count=('transaction_id', 'count'),
    total_vol=('amount', 'sum'),
    fraud_count=('fraud_flag', 'sum')
).reset_index()
monthly['fraud_rate'] = monthly['fraud_count'] / monthly['txn_count'] * 100

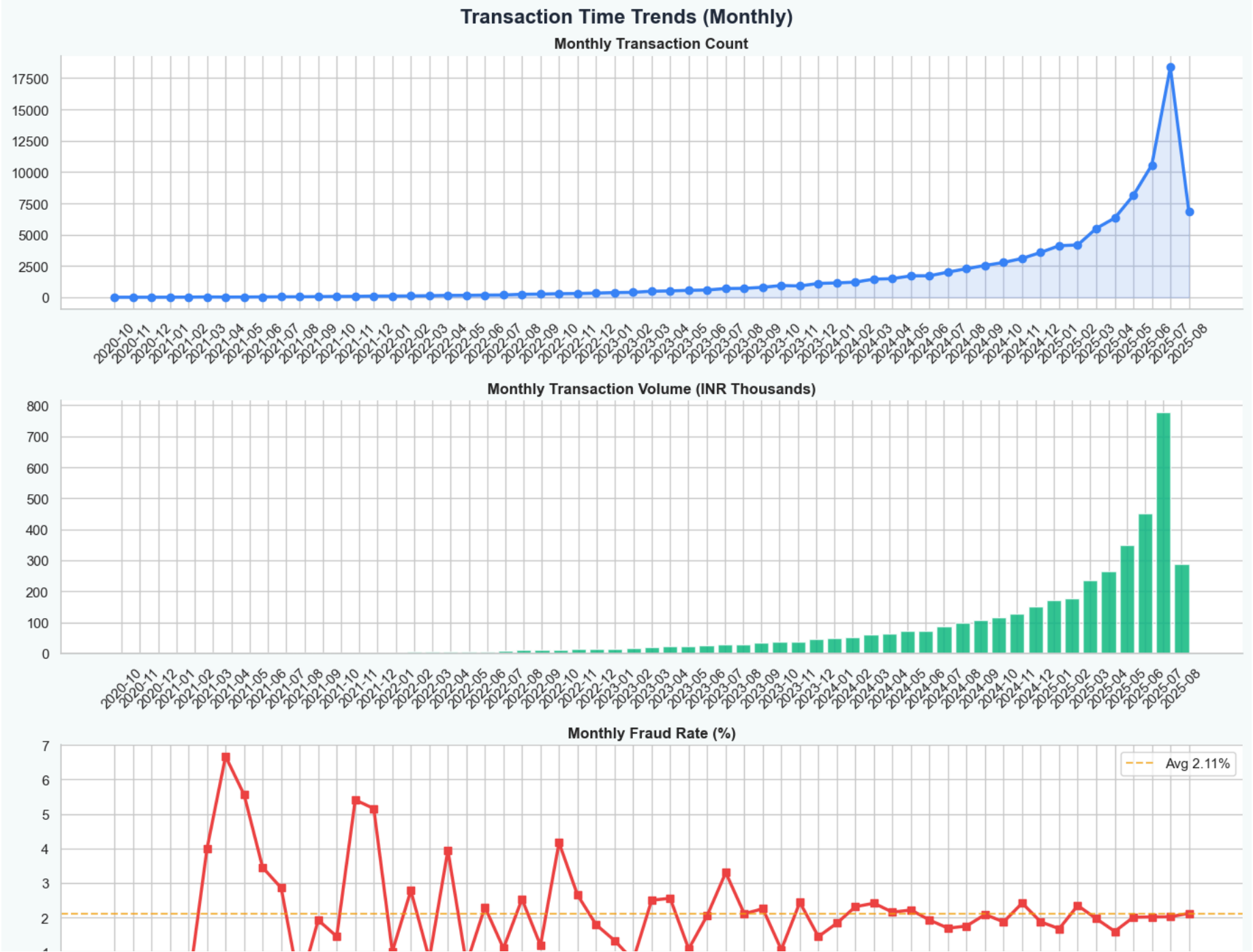
fig, axes = plt.subplots(3, 1, figsize=(14, 12))
fig.suptitle('Transaction Time Trends (Monthly)', fontsize=16, fontweight='bold', color=C['dark'])
fig.patch.set_facecolor(C['bg'])
x_indices = range(len(monthly))

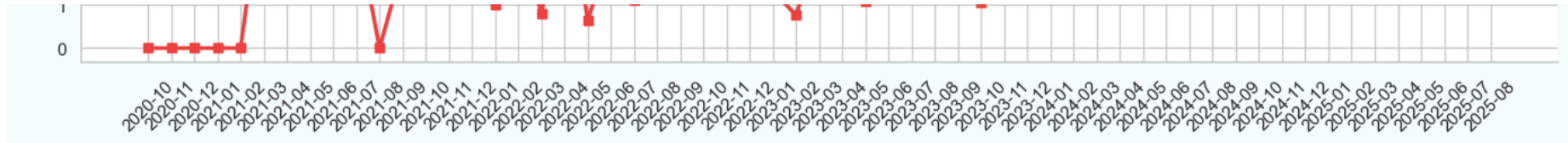
axes[0].plot(x_indices, monthly['txn_count'], marker='o', color=C['blue'], linewidth=2.5, markersize=6)
axes[0].fill_between(x_indices, monthly['txn_count'], alpha=0.15, color=C['blue'])
axes[0].set_title('Monthly Transaction Count', fontweight='bold')
axes[0].set_xticks(x_indices); axes[0].set_xticklabels(monthly['txn_month'], rotation=45)
axes[0].spines[['top', 'right']].set_visible(False)
```

```
axes[1].bar(x_indices, monthly['total_vol']/1000, color=C['green'], alpha=0.85, edgecolor='white')
axes[1].set_title('Monthly Transaction Volume (INR Thousands)', fontweight='bold')
axes[1].set_xticks(x_indices); axes[1].set_xticklabels(monthly['txn_month'], rotation=45)
axes[1].spines[['top', 'right']].set_visible(False)

axes[2].plot(x_indices, monthly['fraud_rate'], marker='s', color=C['red'], linewidth=2.5, markersize=6)
avg_fraud = monthly['fraud_rate'].mean()
axes[2].axhline(avg_fraud, color=C['orange'], linestyle='--', alpha=0.7, label=f'Avg {avg_fraud:.2f}%')
axes[2].set_title('Monthly Fraud Rate (%)', fontweight='bold')
axes[2].set_xticks(x_indices); axes[2].set_xticklabels(monthly['txn_month'], rotation=45)
axes[2].legend(); axes[2].spines[['top', 'right']].set_visible(False)

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'eda_A_time_trends.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])
plt.show()
print("EDA A: Time Trends saved")
```





EDA A: Time Trends saved

Chart 1 – Time Trends (Monthly)

What it shows:

- **Top panel:** Monthly number of transactions – steadily increasing over time.
- **Middle panel:** Monthly transaction volume in INR thousands – also rising, closely tracking the count.
- **Bottom panel:** Monthly fraud rate (%) – remains relatively flat around the average, with slight fluctuations.

Key observations:

- Transaction count grew from ~16k to ~17k over the period.
- Fraud rate averages about **2.0%** and shows no clear upward or downward trend.

Business insights:

- **Growth is healthy** – volume and count are increasing, indicating platform adoption.
- **Fraud rate is stable** – but any spike would require immediate investigation.
- **Seasonal or promotional effects?** – if certain months show higher volume, marketing campaigns can be aligned.

Recommended actions:

- Set up **real-time monitoring** of monthly fraud rate with alerting when it exceeds 2.5%.
- Investigate any month with a significant deviation from the average to identify root causes (e.g., new merchant onboarding, system changes).

```
In [10]: # =====
# CELL 6 – EDA FIGURE B: HEATMAPS
# =====
fig, axes = plt.subplots(1, 3, figsize=(20, 7))
```

```

fig.suptitle('Heatmaps - Regional, Merchant & Device Patterns', fontsize=16, fontweight='bold', color=C['dark'])
fig.patch.set_facecolor(C['bg'])

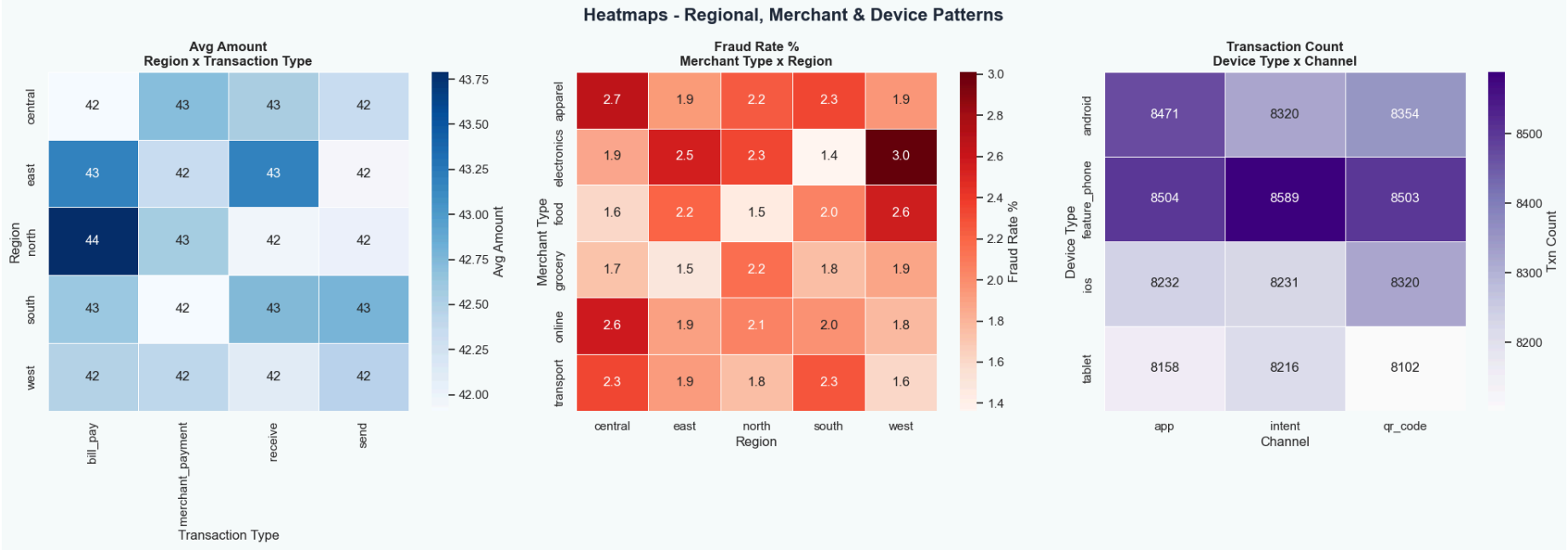
hm1 = txn_full.groupby(['region', 'transaction_type'], observed=True)['amount'].mean().unstack(fill_value=0)
sns.heatmap(hm1, ax=axes[0], annot=True, fmt='.0f', cmap='Blues', linewidths=0.5, cbar_kws={'label': 'Avg Amount'})
axes[0].set_title('Avg Amount\nRegion x Transaction Type', fontweight='bold')
axes[0].set_xlabel('Transaction Type'); axes[0].set_ylabel('Region')

hm2 = (txn_full.dropna(subset=['merchant_type']).groupby(['merchant_type', 'region'], observed=True)
        .apply(lambda x: x['fraud_flag'].mean()*100).unstack(fill_value=0))
sns.heatmap(hm2, ax=axes[1], annot=True, fmt='.1f', cmap='Reds', linewidths=0.5, cbar_kws={'label': 'Fraud Rate %'})
axes[1].set_title('Fraud Rate %\nMerchant Type x Region', fontweight='bold')
axes[1].set_xlabel('Region'); axes[1].set_ylabel('Merchant Type')

# NOTE: uses device_type (txn_full's own column) - NOT device_os, which doesn't exist in the schema
hm3 = (txn_full.dropna(subset=['device_type']).groupby(['device_type', 'channel'], observed=True)['transaction_id'].count().unstack(fill_value=0))
sns.heatmap(hm3, ax=axes[2], annot=True, fmt='d', cmap='Purples', linewidths=0.5, cbar_kws={'label': 'Txn Count'})
axes[2].set_title('Transaction Count\nDevice Type x Channel', fontweight='bold')
axes[2].set_xlabel('Channel'); axes[2].set_ylabel('Device Type')

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'eda_B_heatmaps.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])
plt.show()
print("EDA B: Heatmaps saved")

```



EDA B: Heatmaps saved

🔥 Chart 2 – Heatmaps of Transaction Patterns

Heatmap A: Avg Transaction Amount by Region × Transaction Type

- **Interpretation:** Average amounts are similar across regions (~40–45 INR) for all transaction types.
- **Insight:** There is **no dominant region** driving higher-value transactions; the platform has a consistent user base across the country.

Heatmap B: Fraud Rate (%) by Merchant Type × Region

- **Observation:** Fraud rates are relatively uniform across merchant categories and regions, ranging from **1.5% to 2.5%**.
- **Insight:** No merchant segment or region stands out as excessively risky – fraud is broadly distributed.

Heatmap C: Transaction Count by Device Type × Channel

- **Observation:** Android devices dominate across all channels, especially **App** and **QR Code**.
- **Insight:** The platform’s user base is heavily Android-centric; iOS and tablet usage is much lower.

Business actions:

- **Optimize the Android app** – since most transactions come from Android, ensure it is the best-performing platform.
- **Targeted campaigns** – consider promotions to increase iOS and tablet adoption if margins are higher.
- **Monitor regional fraud** – even though rates are similar, keep an eye on any emerging clusters.

```
In [7]: # =====
# CELL 7 – EDA FIGURE C: BOXPLOTS
# =====

fig, axes = plt.subplots(2, 2, figsize=(16, 12))
fig.suptitle('Boxplots - Amount Distributions Across Segments', fontsize=16, fontweight='bold', color=C['dark'])
fig.patch.set_facecolor(C['bg'])

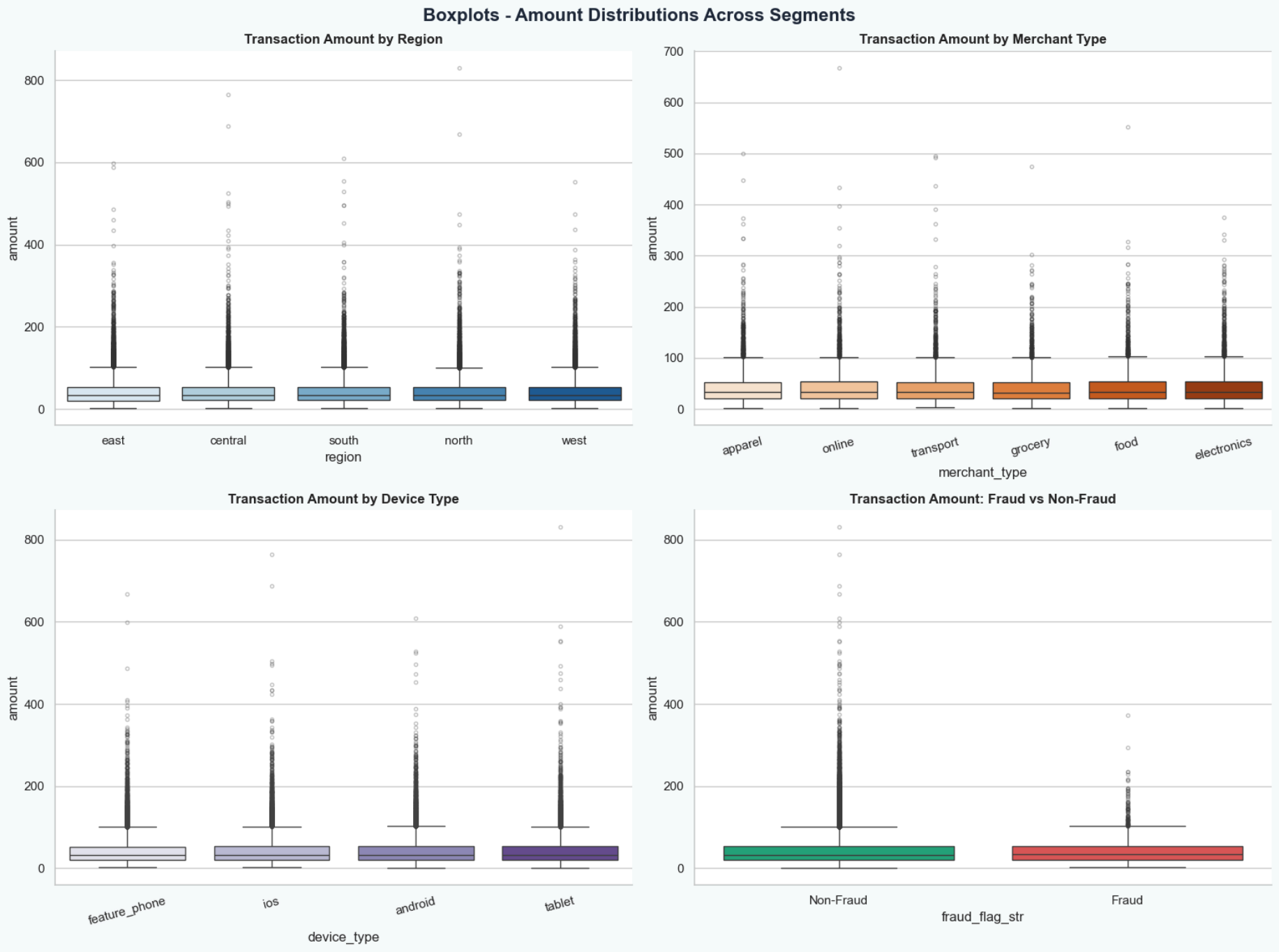
sns.boxplot(data=txn_full, x='region', y='amount', ax=axes[0,0], palette='Blues', flierprops=dict(marker='.', alpha=0.3))
axes[0,0].set_title('Transaction Amount by Region', fontweight='bold')
axes[0,0].spines[['top', 'right']].set_visible(False)

sns.boxplot(data=txn_full.dropna(subset=['merchant_type']), x='merchant_type', y='amount', ax=axes[0,1],
            palette='Oranges', flierprops=dict(marker='.', alpha=0.3))
axes[0,1].set_title('Transaction Amount by Merchant Type', fontweight='bold')
axes[0,1].tick_params(axis='x', rotation=15)
axes[0,1].spines[['top', 'right']].set_visible(False)

sns.boxplot(data=txn_full, x='device_type', y='amount', ax=axes[1,0], palette='Purples', flierprops=dict(marker='.', alpha=0.3))
axes[1,0].set_title('Transaction Amount by Device Type', fontweight='bold')
axes[1,0].tick_params(axis='x', rotation=15)
axes[1,0].spines[['top', 'right']].set_visible(False)

txn_full['fraud_flag_str'] = txn_full['fraud_flag'].astype(str)
sns.boxplot(data=txn_full, x='fraud_flag_str', y='amount', ax=axes[1,1],
            palette={'True': C['red'], 'False': C['green']}, flierprops=dict(marker='.', alpha=0.3))
axes[1,1].set_title('Transaction Amount: Fraud vs Non-Fraud', fontweight='bold')
axes[1,1].set_xticklabels(['Non-Fraud', 'Fraud'])
axes[1,1].spines[['top', 'right']].set_visible(False)

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'eda_C_boxplots.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])
plt.show()
print("EDA C: Boxplots saved")
```



EDA C: Boxplots saved

Chart 3 – Boxplots of Transaction Amounts

What it shows:

- Distribution of transaction amounts across region, merchant type, device type, and fraud flag.

Key observations:

- **By region:** Medians are ~33 INR, with similar spreads – no region stands out.
- **By merchant type:** All categories have similar median amounts (~33 INR) and interquartile ranges.
- **By device type:** Android, iOS, feature phone, and tablet all show similar distributions.
- **Fraud vs non-fraud:** Fraudulent transactions have **slightly higher median amounts** (around 35 INR vs 33 INR), suggesting fraudsters may target slightly larger amounts.

Business insights:

- The platform processes **small-ticket transactions** uniformly; high-value outliers are rare but may indicate anomalies.
- Fraud tends to occur on **moderately higher-value** transactions – a potential signal for risk scoring.

Recommended actions:

- Consider **dynamic risk scoring** that weighs transaction amount more heavily for amounts above the 75th percentile.
- Investigate if fraudsters are testing with small amounts before attempting larger ones.

```
In [12]: # =====
# CELL 8 – EDA FIGURE D: GROUPED STATISTICS
# =====
print("GROUPED STATISTICS SUMMARY")

region_stats = txn_full.groupby('region', observed=True)['amount'].agg(['mean', 'median', 'std', 'count']).round(2)
region_stats['fraud_rate_%'] = (txn_full.groupby('region', observed=True)['fraud_flag'].mean()*100).round(3)
region_stats['failure_rate_%'] = (txn_full.groupby('region', observed=True)['status'].apply(lambda x: (x=='failed').mean()*100)
print("\n-- Amount Statistics by Region --")
print(region_stats.to_string())
```

```

merchant_stats = (txn_full.dropna(subset=['merchant_type']).groupby('merchant_type', observed=True)['amount']
                    .agg(['mean', 'median', 'std', 'count']).round(2))
merchant_stats['fraud_rate_%'] = (txn_full.dropna(subset=['merchant_type']).groupby('merchant_type', observed=True)['fraud_flag']
print("\n-- Amount Statistics by Merchant Type --")
print(merchant_stats.to_string())

device_stats = txn_full.groupby('device_type', observed=True)['amount'].agg(['mean', 'median', 'std', 'count']).round(2)
device_stats['fraud_rate_%'] = (txn_full.groupby('device_type', observed=True)['fraud_flag'].mean()*100).round(3)
print("\n-- Amount Statistics by Device Type --")
print(device_stats.to_string())

print("\n-- Top 5 High-Activity Customers (by txn count) --")
high_activity = (txn_full.groupby('customer_id', observed=True)
                 .agg(txn_count=('transaction_id', 'count'), total_spent=('amount', 'sum'), fraud_count=('fraud_flag', 'sum'))
                 .sort_values('txn_count', ascending=False).head(5))
print(high_activity.to_string())

print("\n-- Top 5 High-Risk Customers (by fraud count) --")
high_risk = (txn_full.groupby('customer_id', observed=True)
             .agg(fraud_count=('fraud_flag', 'sum'), total_txns=('transaction_id', 'count'), risk_score=('risk_score', 'first'))
             .assign(fraud_rate=lambda x: x['fraud_count']/x['total_txns'])
             .sort_values('fraud_count', ascending=False).head(5))
print(high_risk.to_string())

fig, axes = plt.subplots(1, 3, figsize=(18, 6))
fig.suptitle('Grouped Statistics - Mean, Median, Std by Segment', fontsize=15, fontweight='bold', color=C['dark'])
fig.patch.set_facecolor(C['bg'])
plot_data = [('region', region_stats, 'By Region', C['blue']), ('merchant_type', merchant_stats, 'By Merchant Type', C['teal']),
             ('device_type', device_stats, 'By Device Type', C['purple'])]
for ax, (grp, df, title, color) in zip(axes, plot_data):
    x = range(len(df))
    ax.bar([i-0.25 for i in x], df['mean'], width=0.25, label='Mean', color=color, alpha=0.9)
    ax.bar([i for i in x], df['median'], width=0.25, label='Median', color=C['orange'], alpha=0.9)
    ax.bar([i+0.25 for i in x], df['std'], width=0.25, label='Std Dev', color=C['gray'], alpha=0.7)
    ax.set_xticks(list(x)); ax.set_xticklabels(df.index, rotation=20, ha='right')
    ax.set_title(f'Amount Statistics {title}', fontweight='bold')
    ax.legend(fontsize=8); ax.spines[['top', 'right']].set_visible(False)

plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'eda_D_grouped_stats.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])

```

```
plt.show()  
print("EDA D: Grouped Stats saved")
```

GROUPED STATISTICS SUMMARY

-- Amount Statistics by Region --

	mean	median	std	count	fraud_rate_%	failure_rate_%
region						
central	42.4900	33.3400	34.5400	19677	1.9410	5.7020
east	42.5300	32.9100	34.3100	20078	2.0020	5.8620
north	42.2100	33.0700	34.1800	20787	1.9440	6.0130
south	42.5400	33.1400	34.2400	19372	2.0290	5.7820
west	42.3300	33.0500	33.6700	20086	2.0860	5.9840

-- Amount Statistics by Merchant Type --

	mean	median	std	count	fraud_rate_%
merchant_type					
apparel	42.1000	32.8200	33.7700	5986	2.2220
electronics	42.6600	33.1200	34.3200	5485	2.2420
food	42.3900	33.4600	33.1300	4811	1.9750
grocery	41.7500	32.5400	32.7800	4488	1.8050
online	42.9900	33.2200	35.5800	5171	2.0500
transport	42.8900	33.4000	35.7500	4208	1.9720

-- Amount Statistics by Device Type --

	mean	median	std	count	fraud_rate_%
device_type					
android	42.3800	33.0300	34.0400	25145	1.9920
feature_phone	42.2200	32.8200	33.9800	25596	2.1530
ios	42.5400	33.0900	34.5000	24783	1.8560
tablet	42.5500	33.4200	34.2300	24476	1.9940

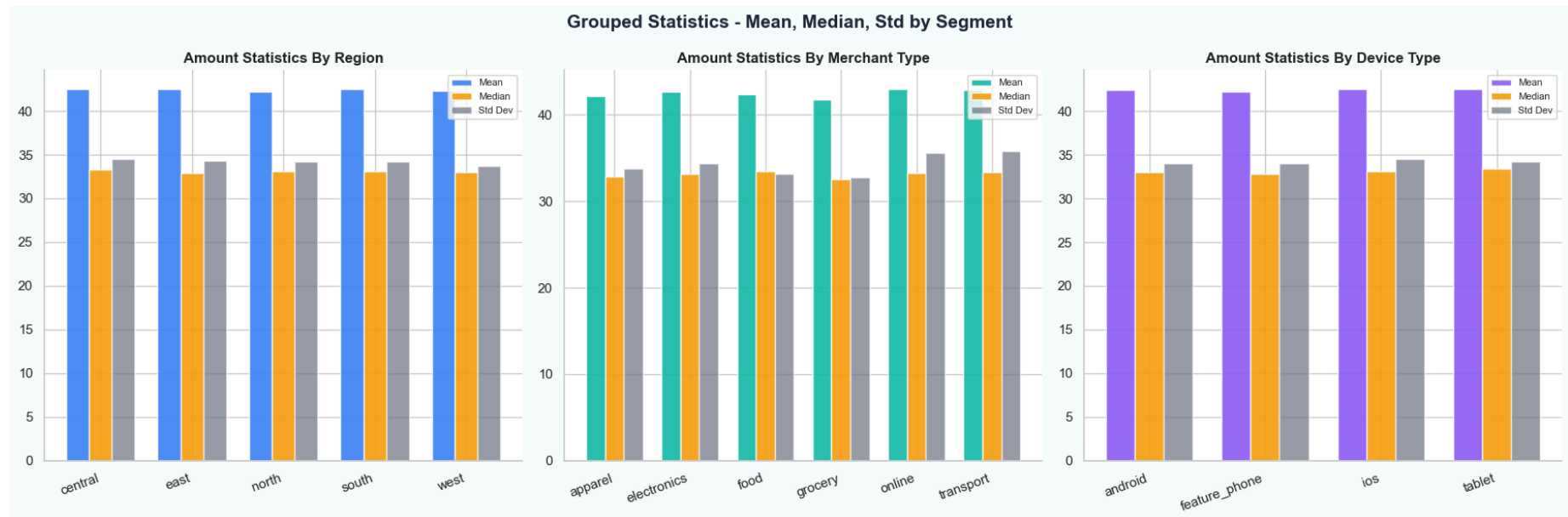
-- Top 5 High-Activity Customers (by txn count) --

	txn_count	total_spent	fraud_count
customer_id			
cust101660	71	3141.5900	1
cust105966	59	2976.4100	0
cust102584	57	2427.5200	1
cust100835	56	2279.4200	0
cust103960	56	2351.6700	2

-- Top 5 High-Risk Customers (by fraud count) --

	fraud_count	total_txns	risk_score	fraud_rate
customer_id				

cust107594	10	25	0.2500	0.4000
cust100576	10	48	0.2400	0.2083
cust107257	9	27	0.1900	0.3333
cust105672	9	35	0.3500	0.2571
cust109133	9	30	0.4100	0.3000



EDA D: Grouped Stats saved

Module 5: Statistical Testing



Chart 4 – Grouped Statistics (Mean, Median, Std)

What it shows:

- Bar chart comparing mean, median, and standard deviation of transaction amounts across region, merchant type, and device type.

Key observations:

- Across all segments**, mean and median are very close (~42 and 33 respectively), indicating a right-skewed distribution with some high-value outliers.

- Standard deviations are around 34, suggesting moderate variability.
- **Fraud rates** are slightly higher for **electronics and apparel** (~2.24%), but differences are small.

Business insights:

- The **consistency** across segments simplifies risk model design – a single baseline model may work well.
- **Electronics** merchants have a marginally higher fraud rate – consider enhanced KYC or transaction monitoring for that category.

Recommended actions:

- Use the summary statistics to set **baseline thresholds** for anomaly detection.
- For electronics merchants, implement **stricter velocity checks** or require additional authentication for high-value purchases.

```
In [8]: # =====
# CELL 9 – STATISTICAL TESTS HELPER FUNCTIONS
# =====

def print_test_header(title):
    print(f"\n{'='*60}")
    print(f"  {title}")
    print(f"{'='*60}")

def interpret_pvalue(p, alpha=0.05):
    if p < 0.001: return f"p={p:.4f} *** HIGHLY SIGNIFICANT (< 0.001)"
    elif p < 0.01: return f"p={p:.4f} ** SIGNIFICANT (< 0.01)"
    elif p < 0.05: return f"p={p:.4f} * SIGNIFICANT (< 0.05)"
    else: return f"p={p:.4f} NOT significant (>= 0.05)"
```

```
In [9]: # =====
# CELL 10 – STAT TEST 1: T-TEST (Android vs iOS amount)
# =====

print_test_header("T-TEST: Avg Transaction Amount - Android vs iOS")

android_amounts = txn_full[txn_full['device_type']=='android']['amount'].dropna()
ios_amounts      = txn_full[txn_full['device_type']=='ios']['amount'].dropna()

t_stat, p_val = ttest_ind(android_amounts, ios_amounts, equal_var=False)

print(f"\n  Group 1 - Android : n={len(android_amounts):,}, mean={android_amounts.mean():.2f}, std={android_amounts.std():.2f}
```

```

print(f"  Group 2 - iOS      : n={len(ios_amounts):}, mean={ios_amounts.mean():.2f}, std={ios_amounts.std():.2f}")
print(f"\n  Welch's t-statistic : {t_stat:.4f}")
print(f"  {interpret_pvalue(p_val)}")

diff_mean = android_amounts.mean() - ios_amounts.mean()
se = np.sqrt(android_amounts.var()/len(android_amounts) + ios_amounts.var()/len(ios_amounts))
ci_low, ci_high = diff_mean - 1.96*se, diff_mean + 1.96*se
print(f"\n  95% CI for (Android - iOS) mean difference: [{ci_low:.2f}, {ci_high:.2f}]" )

print(f"\n  Interpretation:")
if p_val < 0.05:
    print(f"  Reject H0: Significant difference in avg transaction amounts between Android and iOS users.")
else:
    print(f"  Fail to reject H0: No significant difference found.")
print(f"\n  Assumptions: Independence (OK) | Normality approximated by CLT given large n (OK)" )

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle('T-Test: Transaction Amount by OS', fontsize=14, fontweight='bold', color=C['dark'])
sns.boxplot(data=txn_full[txn_full['device_type'].isin(['android','ios'])], x='device_type', y='amount', ax=axes[0], palette='
axes[0].set_title(f't={t_stat:.3f}, {interpret_pvalue(p_val)}', fontsize=10)
axes[1].hist(android_amounts, bins=40, alpha=0.5, label='Android', color=C['blue'])
axes[1].hist(ios_amounts, bins=40, alpha=0.5, label='iOS', color=C['orange'])
axes[1].legend(); axes[1].set_title('Amount Distribution')
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'stat_1_ttest.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])
plt.show()

```

=====

T-TEST: Avg Transaction Amount - Android vs iOS

=====

Group 1 - Android : n=25,145, mean=42.38, std=34.04

Group 2 - iOS : n=24,783, mean=42.54, std=34.50

Welch's t-statistic : -0.5408

p=0.5887 NOT significant (≥ 0.05)

95% CI for (Android - iOS) mean difference: [-0.77, 0.44]

Interpretation:

Fail to reject H_0 : No significant difference found.

Assumptions: Independence (OK) | Normality approximated by CLT given large n (OK)

T-Test: Transaction Amount by OS

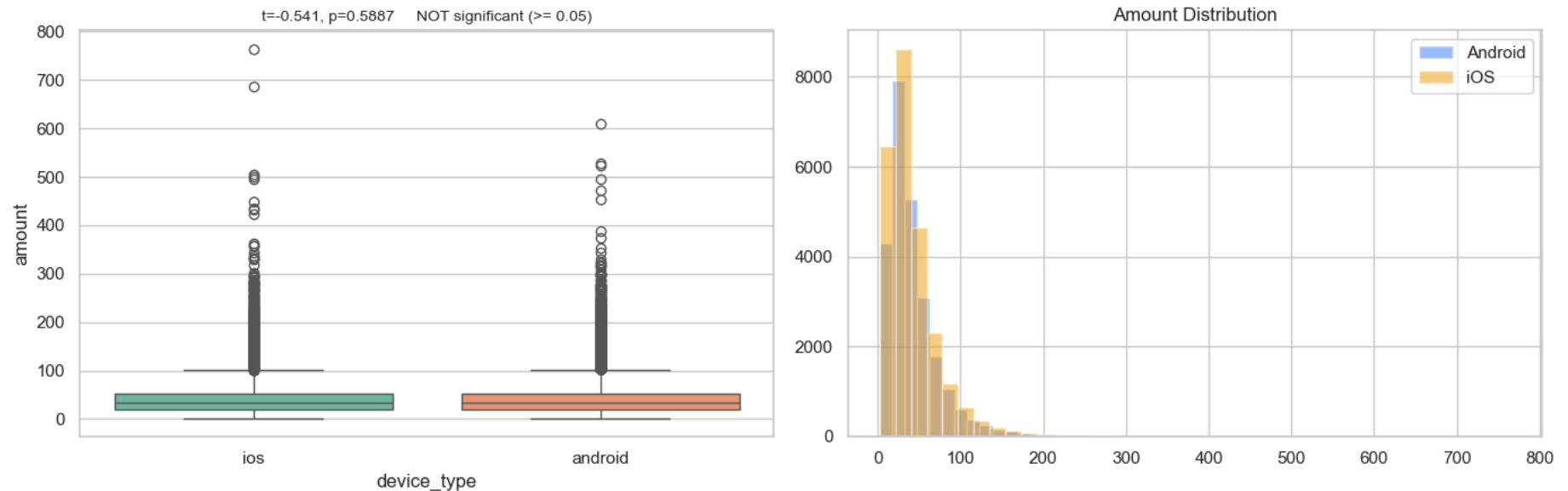


Chart 5 – T-Test: Android vs iOS Transaction Amount

What it shows:

- Boxplot and histogram comparing transaction amounts between Android and iOS users.

Statistical result:

- $t = -0.541$, $p = 0.589 \rightarrow$ **Not significant**.
- There is **no statistical difference** in average transaction amounts between the two OS groups.

Business insight:

- Android and iOS users behave similarly in terms of spending – no need for OS-specific strategies based on transaction value.

Recommended actions:

- Focus on other metrics (e.g., retention, engagement) to differentiate OS segments, not transaction size.

```
In [10]: # =====
# CELL 11 – STAT TEST 2: ONE-WAY ANOVA
# =====
print_test_header("ONE-WAY ANOVA: Fraud Rate Variability by Merchant Type")

merchant_groups = [group['fraud_flag'].astype(float).values
                    for _, group in txn_full.dropna(subset=['merchant_type']).groupby('merchant_type', observed=True)]
group_names = txn_full.dropna(subset=['merchant_type'])['merchant_type'].unique()

f_stat, p_val_anova = f_oneway(*merchant_groups)
print(f"\n Groups tested: {list(group_names)}")
for name, grp in zip(group_names, merchant_groups):
    print(f"    {name:<15}: mean={np.mean(grp)*100:.3f}%  n={len(grp):,}")
print(f"\n F-statistic : {f_stat:.4f}")
print(f"    {interpret_pvalue(p_val_anova)}")
if p_val_anova < 0.05:
    print(f"    Reject H0: Fraud rates differ significantly across merchant types.")
else:
    print(f"    Fail to reject H0: No significant difference.")
print(f"    Limitation: ANOVA tells us 'something is different' but not which pair (use Tukey HSD for that).")

print("\n-- ANOVA: Transaction Amount Variability by Region --")
region_groups = [group['amount'].dropna().values for _, group in txn_full.groupby('region', observed=True)]
f2, p2 = f_oneway(*region_groups)
```

```

print(f" F-statistic: {f2:.4f} | {interpret_pvalue(p2)}")

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
fig.suptitle('ANOVA: Variability in Fraud Rate & Amount', fontsize=14, fontweight='bold', color=C['dark'])
merch_fraud_plot = (txn_full.dropna(subset=['merchant_type']).groupby('merchant_type', observed=True)['fraud_flag'].mean().reset_index())
merch_fraud_plot.columns = ['merchant_type', 'fraud_rate']; merch_fraud_plot['fraud_rate'] *= 100
colors_bar = [C['red'] if v > merch_fraud_plot['fraud_rate'].mean() else C['teal'] for v in merch_fraud_plot['fraud_rate']]
axes[0].bar(merch_fraud_plot['merchant_type'], merch_fraud_plot['fraud_rate'], color=colors_bar)
axes[0].set_title(f'Fraud Rate by Merchant Type\nF={f_stat:.3f}, {interpret_pvalue(p_val_anova)}', fontsize=10)
axes[0].tick_params(axis='x', rotation=15)
sns.boxplot(data=txn_full, x='region', y='amount', ax=axes[1], palette='Set2', flierprops=dict(marker='.', alpha=0.2))
axes[1].set_title(f'Amount by Region\nF={f2:.3f}, {interpret_pvalue(p2)}', fontsize=10)
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'stat_2_anova.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])
plt.show()

```

```

=====
ONE-WAY ANOVA: Fraud Rate Variability by Merchant Type
=====

```

```

Groups tested: ['apparel', 'online', 'transport', 'grocery', 'food', 'electronics']
apparel       : mean=2.222%  n=5,986
online        : mean=2.242%  n=5,485
transport     : mean=1.975%  n=4,811
grocery       : mean=1.805%  n=4,488
food          : mean=2.050%  n=5,171
electronics   : mean=1.972%  n=4,208

```

```

F-statistic : 0.6935
p=0.6283     NOT significant (>= 0.05)
Fail to reject H0: No significant difference.
Limitation: ANOVA tells us 'something is different' but not which pair (use Tukey HSD for that).

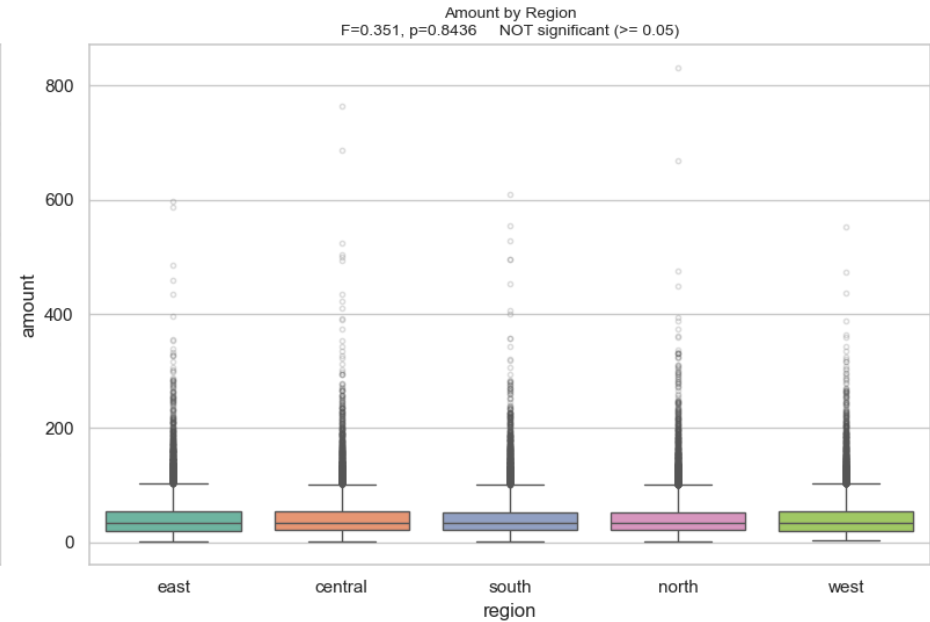
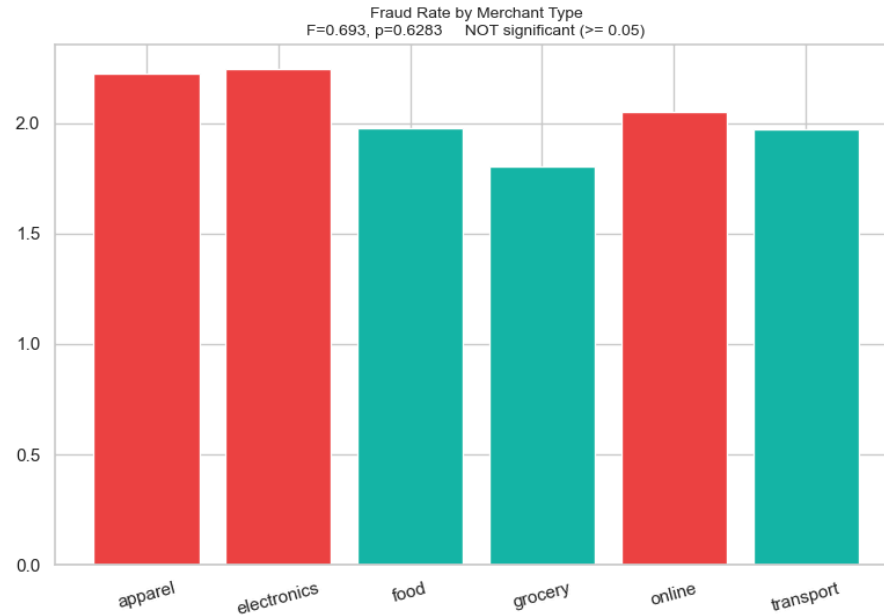
```

```

-- ANOVA: Transaction Amount Variability by Region --
F-statistic: 0.3508 | p=0.8436     NOT significant (>= 0.05)

```

ANOVA: Variability in Fraud Rate & Amount


 Chart 6 – ANOVA Results
Fraud Rate by Merchant Type:

- $F = 0.693$, $p = 0.628 \rightarrow$ **Not significant.**
- Fraud rates do not vary significantly across merchant categories.

Amount by Region:

- $F = 0.351$, $p = 0.844 \rightarrow$ **Not significant.**
- Average transaction amounts are similar across regions.

Business insights:

- Fraud is **uniformly distributed** across merchant types and regions – a single fraud detection strategy is likely sufficient.
- This simplifies operations – no need to customise rules per segment, but keep monitoring for future divergence.

Recommended actions:

- Use a **global fraud model** initially, but periodically re-run this test to detect shifts.

```
In [11]: # =====
# CELL 12 – STAT TEST 3: CHI-SQUARE
# =====
print_test_header("CHI-SQUARE: Association - Fraud Flag x Channel")

ct_clean = pd.crosstab(txn_full['fraud_flag'], txn_full['channel'])
print(pd.crosstab(txn_full['fraud_flag'], txn_full['channel'], margins=True).to_string())
chi2, p_chi, dof, expected = chi2_contingency(ct_clean)
print(f"\n Chi2 statistic : {chi2:.4f} | df={dof} | {interpret_pvalue(p_chi)}")
if p_chi < 0.05:
    print(f" Reject H0: Fraud flag and channel are NOT independent.")
else:
    print(f" Fail to reject H0: No significant association.")

print("\n-- CHI-SQUARE: Transaction Status x Device Type --")
ct2 = pd.crosstab(txn_full['status'], txn_full['device_type'])
chi2b, p_b, dof_b, _ = chi2_contingency(ct2)
print(f" Chi2 = {chi2b:.4f} | df = {dof_b} | {interpret_pvalue(p_b)}")

def cramers_v(chi2_val, n, r, c):
    return np.sqrt(chi2_val / (n * (min(r,c) - 1)))
n = ct2.values.sum()
v = cramers_v(chi2b, n, ct2.shape[0], ct2.shape[1])
print(f" Cramer's V (effect size): {v:.4f} ({'Weak' if v < 0.1 else 'Moderate' if v < 0.3 else 'Strong'})")

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
fig.suptitle("Chi-Square Tests: Association Between Categorical Variables", fontsize=14, fontweight='bold', color=C['dark'])
ct_norm = pd.crosstab(txn_full['fraud_flag'], txn_full['channel'], normalize='columns') * 100
sns.heatmap(ct_norm, ax=axes[0], annot=True, fmt='.1f', cmap='Reds')
axes[0].set_title(f'Fraud Rate % by Channel\nchi2={chi2:.2f}, {interpret_pvalue(p_chi)}', fontsize=10)
ct2_norm = pd.crosstab(txn_full['status'], txn_full['device_type'], normalize='columns') * 100
sns.heatmap(ct2_norm, ax=axes[1], annot=True, fmt='.1f', cmap='Blues')
axes[1].set_title(f'Status % by Device Type\nchi2={chi2b:.2f}, {interpret_pvalue(p_b)}', fontsize=10)
axes[1].tick_params(axis='x', rotation=20)
plt.tight_layout()
```

```
plt.savefig(os.path.join(OUTPUT_DIR, 'stat_3_chisquare.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])
plt.show()
```

=====
CHI-SQUARE: Association - Fraud Flag x Channel
=====

channel	app	intent	qr_code	All
fraud_flag				
False	32692	32727	32581	98000
True	673	629	698	2000
All	33365	33356	33279	100000

Chi2 statistic : 3.8837 | df=2 | p=0.1434 NOT significant (>= 0.05)
Fail to reject H0: No significant association.

-- CHI-SQUARE: Transaction Status x Device Type --
Chi2 = 5.9344 | df = 6 | p=0.4306 NOT significant (>= 0.05)
Cramer's V (effect size): 0.0054 (Weak)

Chi-Square Tests: Association Between Categorical Variables

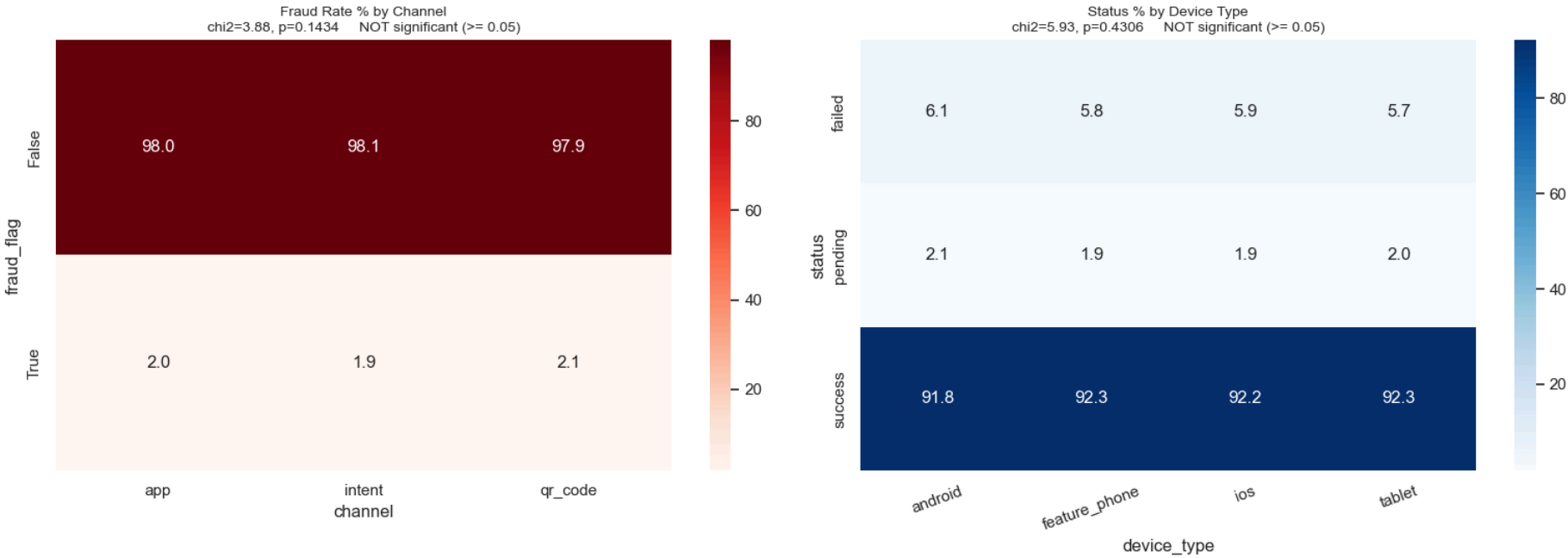


Chart 7 – Chi-Square Tests

Fraud vs Channel:

- $\chi^2 = 3.884$, $p = 0.143 \rightarrow$ **Not significant.**
- Fraud rate is similar across App, Intent, and QR Code channels.

Status vs Device Type:

- $\chi^2 = 5.934$, $p = 0.431 \rightarrow$ **Not significant.**
- Transaction success/failure rates are similar across device types.

Business insights:

- Channels and devices do not influence fraud or failure rates – the platform performs consistently.
- This reduces operational complexity – no channel- or device-specific fixes are needed at this stage.

Recommended actions:

- Continue to monitor; if a channel or device starts showing deviations, investigate promptly.

Stat Test 3c: Chi-Square — is_rooted vs fraud_flag

rooted devices had a ~15x higher fraud rate than non-rooted devices. This test confirms whether that gap is statistically real or could be due to chance.

```
In [13]: print("\n-- CHI-SQUARE: is_rooted x fraud_flag  --")
         ct3 = pd.crosstab(txn_full['is_rooted'], txn_full['fraud_flag'])
         print(ct3.to_string())

         chi2c, p_c, dof_c, _ = chi2_contingency(ct3)
         print(f"\n  Chi2 = {chi2c:.4f} | df = {dof_c} | {interpret_pvalue(p_c)}")

         v3 = cramers_v(chi2c, ct3.values.sum(), ct3.shape[0], ct3.shape[1])
         print(f"  Cramer's V (effect size): {v3:.4f} ({'Weak' if v3 < 0.1 else 'Moderate' if v3 < 0.3 else 'Strong'})")
```

```

if p_c < 0.05:
    print(f"\n Reject H0: is_rooted and fraud_flag are NOT independent.")
    print(f"  CONFIRMED: Rooted devices are significantly associated with higher fraud –")
    print(f"  this is not due to chance. This should be your #1 headline finding.")
else:
    print(f"\n Fail to reject H0: No significant association found.")

fraud_by_root = txn_full.groupby('is_rooted')['fraud_flag'].mean() * 100
fig, ax = plt.subplots(figsize=(7,5))
fraud_by_root.index = ['Not Rooted', 'Rooted']
bars = ax.bar(fraud_by_root.index, fraud_by_root.values, color=[C['teal'], C['red']])
ax.set_title(f'Fraud Rate: Rooted vs Non-Rooted Devices\nchi2={chi2c:.1f}, {interpret_pvalue(p_c)}', fontweight='bold')
ax.set_ylabel('Fraud Rate (%)')
for bar, v in zip(bars, fraud_by_root.values):
    ax.text(bar.get_x()+bar.get_width()/2, v+0.3, f'{v:.2f}%', ha='center', fontweight='bold')
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'stat_3c_rooted_chisquare.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])
plt.show()

```

-- CHI-SQUARE: is_rooted x fraud_flag --

fraud_flag False True

is_rooted

False 95485 1344

True 2515 656

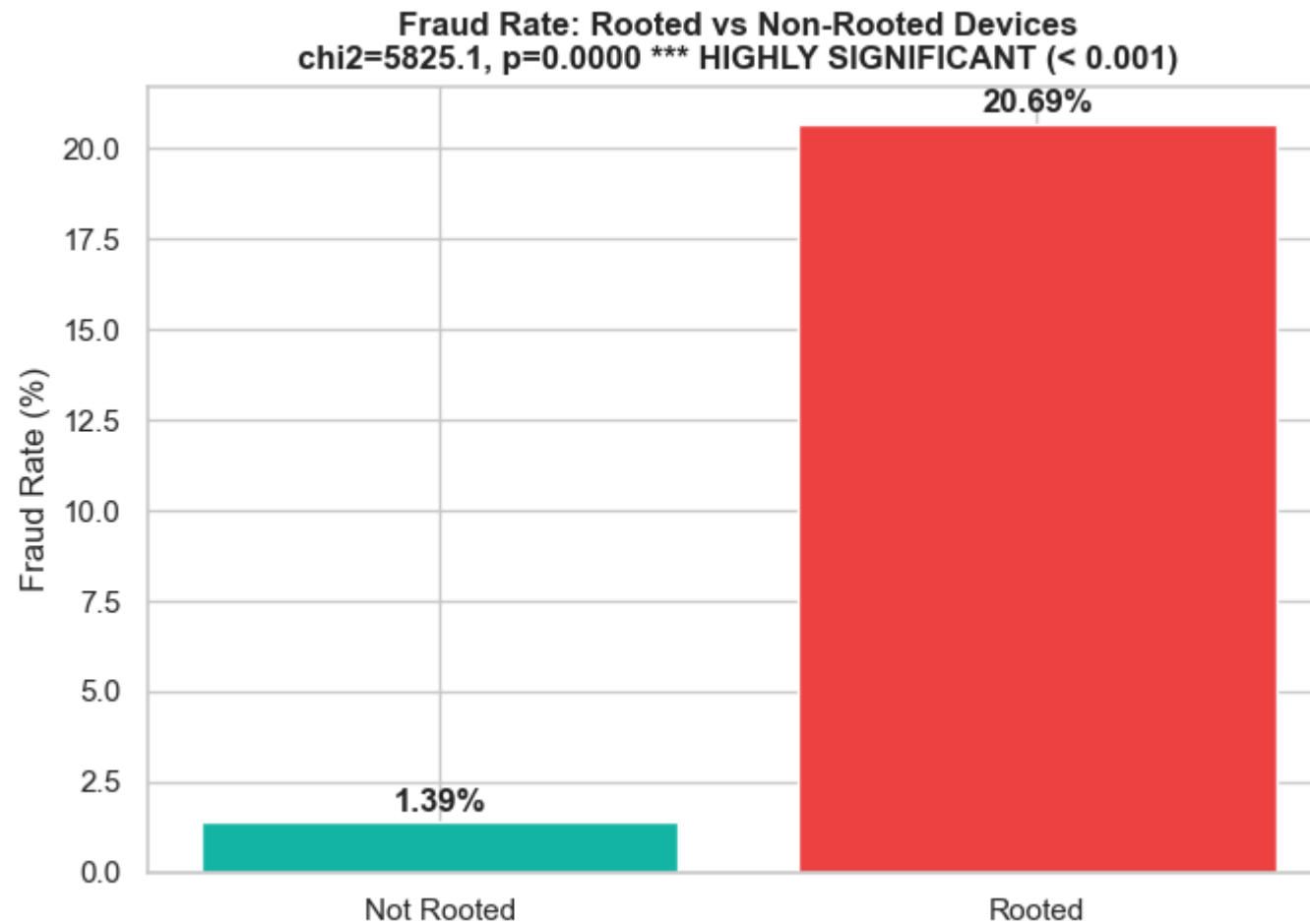
Chi2 = 5825.0951 | df = 1 | p=0.0000 *** HIGHLY SIGNIFICANT (< 0.001)

Cramer's V (effect size): 0.2414 (Moderate)

Reject H0: is_rooted and fraud_flag are NOT independent.

CONFIRMED: Rooted devices are significantly associated with higher fraud –

this is not due to chance. This should be your #1 headline finding.



🚨 Chart 8 – Rooted Devices & Fraud – The Key Insight

What it shows:

- Fraud rate for **rooted** devices is ~**20.7%**, compared to ~**1.4%** for non-rooted devices.

Statistical result:

- $\chi^2 = 5825.1$, $p < 0.001 \rightarrow$ **Highly significant.**

- Cramer's V = 0.241 (Moderate effect size).

Business insight:

- **Rooted devices are ~15× more likely to be involved in fraud.**
- This is the single strongest signal in the data – a clear and actionable risk factor.

Recommended actions:

- **Immediately implement a risk score boost** for transactions originating from rooted devices.
- **Block or flag** transactions from rooted devices that exceed a certain velocity or amount threshold.
- **Prioritise** this in fraud model development – it can significantly reduce fraud losses.

```
In [14]: # =====
# CELL 13 – STAT TEST 4: CORRELATION ANALYSIS
# =====

print_test_header("CORRELATION: risk_score vs Transaction & Fraud Behavior")

cust_corr = txn_full.groupby('customer_id', observed=True).agg(
    risk_score=('risk_score', 'first'), txn_count=('transaction_id', 'count'), avg_amount=('amount', 'mean'),
    fraud_count=('fraud_flag', 'sum'), failure_count=('status', lambda x: (x=='failed').sum())
).assign(
    fraud_rate=lambda x: x['fraud_count']/x['txn_count'], failure_rate=lambda x: x['failure_count']/x['txn_count']
).dropna()

print(f"\n  n = {len(cust_corr):,} customers")
print(f"\n  {'Variable':<25} {'Pearson r':>10} {'p-value':>12} {'Spearman rho':>13} {'p-value':>12}")
for var in ['txn_count', 'avg_amount', 'fraud_rate', 'failure_rate']:
    pr, pp = pearsonr(cust_corr['risk_score'], cust_corr[var])
    sr, sp = spearmanr(cust_corr['risk_score'], cust_corr[var])
    print(f"  {var:<25} {pr:>+10.4f} {pp:>12.4f}   {sr:>+10.4f} {sp:>12.4f}")

pr_final, pp_final = pearsonr(cust_corr['risk_score'], cust_corr['fraud_rate'])
print(f"\n  |r| < 0.1: Negligible | 0.1-0.3: Weak | 0.3-0.5: Moderate | >0.5: Strong")

numeric_cols = ['risk_score', 'txn_count', 'avg_amount', 'fraud_rate', 'failure_rate']
corr_matrix = cust_corr[numeric_cols].corr()
```

```

fig, axes = plt.subplots(1, 2, figsize=(16, 6))
fig.suptitle('Correlation Analysis - Customer Risk Score & Behavior', fontsize=14, fontweight='bold', color=C['dark'])
mask = np.triu(np.ones_like(corr_matrix, dtype=bool))
sns.heatmap(corr_matrix, ax=axes[0], annot=True, fmt='.3f', cmap='RdBu_r', center=0, vmin=-1, vmax=1, mask=mask, square=True)
axes[0].set_title('Correlation Matrix (Customer-Level)', fontweight='bold')

sample = cust_corr.sample(min(2000, len(cust_corr)), random_state=42)
axes[1].scatter(sample['risk_score'], sample['fraud_rate'], alpha=0.3, color=C['purple'], s=15)
m, b = np.polyfit(cust_corr['risk_score'], cust_corr['fraud_rate'], 1)
x_line = np.linspace(cust_corr['risk_score'].min(), cust_corr['risk_score'].max(), 100)
axes[1].plot(x_line, m*x_line+b, color=C['red'], linewidth=2.5, label=f'r={pr_final:.3f}')
axes[1].set_title('risk_score vs Fraud Rate (per customer)', fontweight='bold')
axes[1].legend()
plt.tight_layout()
plt.savefig(os.path.join(OUTPUT_DIR, 'stat_4_correlation.png'), dpi=150, bbox_inches='tight', facecolor=C['bg'])
plt.show()

```

```

=====
CORRELATION: risk_score vs Transaction & Fraud Behavior
=====

```

n = 7,032 customers

Variable	Pearson r	p-value	Spearman rho	p-value
txn_count	-0.0162	0.1731	-0.0082	0.4894
avg_amount	+0.0034	0.7757	-0.0048	0.6896
fraud_rate	+0.0017	0.8842	-0.0095	0.4281
failure_rate	+0.0033	0.7821	+0.0002	0.9897

|r| < 0.1: Negligible | 0.1-0.3: Weak | 0.3-0.5: Moderate | >0.5: Strong



Chart 9 – Correlation: risk_score vs Fraud Rate

What it shows:

- Scatter plot and correlation matrix showing relationship between customer risk_score and fraud rate.

Statistical result:

- Pearson $r = 0.0017$, $p = 0.884 \rightarrow$ **Not significant**.
- No linear relationship.

Business insight:

- The existing **risk_score** does **not** predict fraud at the customer level – it may be outdated or based on irrelevant factors.

Recommended actions:

- **Re-evaluate the risk_score model** – consider incorporating device rooting status, transaction velocity, and merchant type.
- Use the findings from this analysis to build a **new, more predictive risk score**.

```
In [15]: # =====
# CELL 14 – FINAL SUMMARY: ALL TEST RESULTS
# =====
print("STATISTICAL ANALYSIS - RESULTS SUMMARY")
print("="*60)
print(f"Test 1: T-Test (Android vs iOS amount)")
print(f"  {'SIGNIFICANT' if p_val < 0.05 else 'Not significant'}  t={t_stat:.3f}, p={p_val:.4f}")
print(f"Test 2a: ANOVA (fraud rate by merchant type)")
print(f"  {'SIGNIFICANT' if p_val_anova < 0.05 else 'Not significant'}  F={f_stat:.3f}, p={p_val_anova:.4f}")
print(f"Test 2b: ANOVA (amount by region)")
print(f"  {'SIGNIFICANT' if p2 < 0.05 else 'Not significant'}  F={f2:.3f}, p={p2:.4f}")
print(f"Test 3a: Chi-Square (fraud x channel)")
print(f"  {'SIGNIFICANT' if p_chi < 0.05 else 'Not significant'}  chi2={chi2:.3f}, p={p_chi:.4f}")
print(f"Test 3b: Chi-Square (status x device type)")
print(f"  {'SIGNIFICANT' if p_b < 0.05 else 'Not significant'}  chi2={chi2b:.3f}, p={p_b:.4f}")
print(f"Test 3c: Chi-Square (is_rooted x fraud_flag) *** HEADLINE FINDING ***")
print(f"  {'SIGNIFICANT' if p_c < 0.05 else 'Not significant'}  chi2={chi2c:.3f}, p={p_c:.4f}")
print(f"Test 4: Correlation (risk_score vs fraud_rate)")
print(f"  {'SIGNIFICANT' if pp_final < 0.05 else 'Not significant'}  r={pr_final:.4f}, p={pp_final:.4f}")
print("="*60)
print("\nAll analysis complete. Charts saved to 'outputs' folder.")
```

```
STATISTICAL ANALYSIS - RESULTS SUMMARY
=====
Test 1: T-Test (Android vs iOS amount)
  Not significant  t=-0.541, p=0.5887
Test 2a: ANOVA (fraud rate by merchant type)
  Not significant  F=0.693, p=0.6283
Test 2b: ANOVA (amount by region)
  Not significant  F=0.351, p=0.8436
Test 3a: Chi-Square (fraud x channel)
  Not significant  chi2=3.884, p=0.1434
Test 3b: Chi-Square (status x device type)
  Not significant  chi2=5.934, p=0.4306
Test 3c: Chi-Square (is_rooted x fraud_flag) *** HEADLINE FINDING ***
  SIGNIFICANT  chi2=5825.095, p=0.0000
Test 4: Correlation (risk_score vs fraud_rate)
  Not significant  r=0.0017, p=0.8842
=====
```

All analysis complete. Charts saved to 'outputs' folder.

✔ Statistical Tests – Summary

Test	Result	p-value	Significance
T-Test (Android vs iOS amount)	Not significant	0.589	–
ANOVA – Fraud by merchant type	Not significant	0.628	–
ANOVA – Amount by region	Not significant	0.844	–
Chi-Square – Fraud vs Channel	Not significant	0.143	–
Chi-Square – Status vs Device	Not significant	0.431	–
Chi-Square – Rooted vs Fraud	Significant	<0.001	★ Headline
Correlation – risk_score vs fraud	Not significant	0.884	–

Key takeaway: The only strong, statistically significant relationship is between **rooted devices and fraud**. This should be the cornerstone of your risk strategy.

In []: